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Settlement morphology, demographics, and governance challenges of water, sanitation, and hygiene in low-income urban communities in Ghana

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Abstract

Access to safe water, sanitation, and hygiene (WASH) remains deeply unequal in rapidly urbanizing landscapes of the Global South, underlain by intra-settlement micro-geographies of exclusion, yet remain under-examined. Drawing on mixed-method case study of 150 households of two socio-culturally diverse low-income communities in Ghana's Greater Accra Metropolitan Area, this study integrates household surveys, spatial analysis, and qualitative interviews to quantify how settlement morphology, socio-economic stratification, and the contribution of communities shape WASH access. Results show statistically significant differences in improved water access (75% in Chorkor vs. 56.4% in Nima; $p = 0.017$), yet sanitation deprivation is pervasive in both communities, with 87.2% of households in Chorkor and 91.7% in Nima relying on unimproved facilities with two-thirds of households depending on public toilets (62.8% and 63.9%, respectively), reinforcing shared sanitation burden. Distance to water sources remains significant ($p < 0.05$), with mean fetching distances of 69.49 m in Chorkor and 54.93 m in Nima, while income strongly predicts access to water sources in both sites ($p < 0.001$ in Chorkor; $p = 0.016$ in Nima). Solid waste infrastructure reveals contrasting patterns, with fragmented spatial distribution in Chorkor against more even coverage in Nima where improved access reaches 48.6% compared to 33.3% in Chorkor. Notably, community co-production significantly enhances service outcomes in Nima (water $p = 0.002$; toilets $p = 0.047$; waste $p = 0.002$), but not in Chorkor, indicating differences in governance. These findings reveal two distinct regimes of exclusion of spatially segmented inequality in Chorkor and settlement-wide systemic under-service in Nima. The composite access scoring (0–50) demonstrates that WASH exclusion is not uniform but patterned along spatial, socio-economic, and institutional gradients, demanding context-specific upgrading strategies that integrate spatial justice, affordability, and participatory co-management. Therefore, the findings reinforce the broader argument that WASH inequality is not simply a technical deficit but a manifestation of uneven urban development, political marginalisation, climate vulnerability, and governance fragmentation.

Keywords: Urban poverty, Migrant community, Household vulnerability, Socioeconomic barriers, Community co-production

1 Introduction

Water, sanitation, and hygiene (WASH) services are not only fundamental to public health and well-being but also are integral to achieving social equity and human dignity, remaining as a central pillar of the Sustainable Development Goals (SDGs) [74]. However, despite global commitments to WASH, access to safe water and sanitation remains deeply unequal across the Global South, particularly in low-income urban communities [30, 80]. Forecasts suggest that, without accelerated action, Sub-Saharan Africa will account for the highest global mortality and economic burden linked to poor WASH by 2030 [29], as informal settlements across Africa, Asia, and Latin America continue to exhibit fragmented WASH-related governance and infrastructure [6, 47]. Currently, 2.1–2.2 billion people lack safely managed drinking water, whilst 3.4 billion are lacking safely managed sanitation [80]. The burden is most pronounced in low-income urban areas where rapid growth mostly outpaces infrastructure development, producing stark inequalities in access, and having direct implications for global health, social justice, and economic productivity [11, 13, 33, 53, 79–81]. However, urban WASH research tends to treat "low-income/informal settlements" as internally homogeneous and often rely on binary service categories (improved/unimproved or basic/limited) that obscure within-settlement nuanced variation in affordability, distance, reliability, and sharing burdens. Unsafe WASH conditions globally are linked to waterborne diseases and responsible for over 829,000 deaths annually, particularly among children under five [1, 79].

Additionally, household solid waste poses a growing environmental and public health challenge in rapidly urbanizing cities of the Global South in the face of progress in demographic transition. Population growth and migration have intensified waste generation in low-income and informal settlements, where municipal collection systems often remain inadequate [38, 73]. In Ghanaian cities, uncollected waste frequently accumulates in drains and water bodies, which in turn increase flood risks, vector breeding, and disease transmission [46, 55]. Evidence increasingly shows that WASH co-production through partnerships among municipalities, community groups, and informal waste collectors can improve collection coverage, accountability, and service reliability in underserved areas [25, 33, 44, 47]. Co-production in WASH constitutes an important yet often neglected structure of WASH governance and denotes the joint production of services by citizens and public (and often private) providers, where users contribute labour, resources, monitoring, and decision-making to sustain water and sanitation systems [54]. Urban studies suggest that co-production is common in the Global South, compensating for fragmented formal networks while producing hybrid governance arrangements [25, 47]. Low-income urban neighbourhoods in Ghana's Greater Accra Metropolitan Area (GAMA) exemplifies this reality in the face of unreliable piped water supply, costly dependence on vendors, and extensive reliance on shared sanitation [2, 16, 42, 62]. It is reported that more than 60% of low-income households in GAMA rely on shared sanitation, far exceeding the national average [80]. Affordability and reliability remain core barriers, with poor households paying disproportionately higher fees for water from vendors compared to those with direct piped access [5, 7, 68, 69, 76].

Coastal indigenous low-income settlements and inland migrant-dominated urban communities often experience distinct WASH access and uptake pathways, shaped mostly by livelihood structures, tenure arrangements, and social organization. In

GAMA and comparable cities, studies show that income, education, and neighbourhood identity strongly condition access to improved water and sanitation [5, 42, 67]. Migrant-dominated settlements with high tenancy and density tend to exhibit differentiated service preferences and collective management responses under constrained supply [16, 68, 69], while indigenous coastal communities face livelihood-linked vulnerabilities and spatial marginalisation affecting sanitation uptake [20, 33, 47]. Against this backdrop, the case of GAMA is particularly compelling as it reflects both the localized dynamics of infrastructure deprivation and the broader challenges facing rapidly urbanizing cities in the Global South. Hence, this study aims to analyse how spatial differentiation, sociodemographic context, governance barriers, and communal adaptation shape households' access to WASH facilities in two low-income urban settlements to understand context-sensitive pathways for equitable and resilient WASH service delivery, enabling a nuanced analysis of WASH inequality.

2 Theoretical framework

WASH inequalities in low-income urban communities are best understood as the outcome of interacting socio-spatial and institutional forces rather than as isolated infrastructure deficits. At the structural level, settlement morphology which includes including density, plot irregularity, road width, drainage systems, flood exposure, and tenure insecurity shapes both the technical feasibility and environmental safety of service provision. Engineering and planning research demonstrates that dense and spatially constrained informal settlements limit the extension of sewer networks, complicate faecal sludge management logistics, and reduce the suitability of conventional sanitation technologies [64, 72]. In the Greater Accra Metropolitan Area (GAMA), spatial analyses using census and remote sensing data show that deprived neighbourhoods are characterized by high building congestion and infrastructure deficits that correlate with limited-service provision [41]. Such morphological constraints are further compounded by rapid urbanization and informal land development, which historically outpace coordinated infrastructure investment [45]. However, spatial form alone does not determine WASH outcomes. Demographic dynamics including household size, tenancy status, migration history, income, education, and gendered power relations mediate exposure and vulnerability. Empirical evidence from Accra shows that households in compound housing and tenant arrangements are more likely to rely on shared sanitation and vendor-supplied water, incurring higher per-unit costs and experiencing greater service unreliability [4, 27, 68, 69]. Globally, affordability constraints and service intermittency disproportionately affect lower-income urban residents [14]. When combined, settlement morphology and demographic differentiation operate as upstream determinants that structure who can secure safe, reliable, and affordable WASH services. Neighbourhoods characterized by high density, irregular layouts, limited accessibility, and insecure tenure therefore tend to exhibit significantly lower levels of safely managed water and sanitation than spatially integrated areas [64, 72], while larger, poorer, and tenant households experience higher levels of WASH deprivation and dependence on informal arrangements [27, 68, 69].

This study uses an Integrated Socio-Spatial Governance Framework for Urban WASH Inequality, that is theoretically grounded in, and conceptually derived from, established bodies of literature in urban political ecology [58, 71], institutional governance theory [54, 66], socio-spatial inequality scholarship [45], and settlement morphology analysis [41, 64]. Beyond spatial and demographic determinants, governance and institutional fragmentation critically shape the distribution, regulation, and monitoring of WASH services. Global evidence indicates that institutional overlap, political interference, financing constraints, and information gaps are recurring barriers to effective WASH delivery in informal settlements [66]. In Ghana, sectoral evolution and regulatory fragmentation have produced blurred accountability between utilities, metropolitan assemblies, environmental health units, landlords, and private providers [3]. Studies of sanitation by-law enforcement further reveal weaknesses in monitoring, sanctions, and public compliance within urban jurisdictions [39]. Inadequate neighbourhood-level data reflecting both technical and political exclusion limit evidence-based allocation of resources and reinforces the marginalization of informal settlements in planning systems [41]. These governance deficits foster the emergence of informal service ecologies (Fig. 1), in which households rely on tanker deliveries, sachet water markets, shared/public toilets, and informal desludging services to bridge systemic gaps. While such arrangements provide short-term coping mechanisms, they frequently shift higher financial and health risks onto poorer households through elevated tariffs, unreliable supply, and contamination exposure [15, 58, 68, 69]. These interacting forces collectively conceptualize WASH deprivation in Greater Accra as a co-produced outcome of settlement form, demographic pressures, governance fragmentation, and informal adaptive capacity, demonstrating how socio-spatial inequality becomes materially integrated within urban service systems and perpetuates disparities in access to safely managed water and sanitation (Fig. 1).

WASH is therefore driven by interacting structural, demographic, and institutional forces (Fig. 1). At the macro level, rapid urbanization, poverty, informality, and climate

Fig. 1 Integrated socio-spatial governance framework explaining how WASH inequalities in low-income urban communities of the Greater Accra Metropolitan Area (GAMA), Ghana

risk create the broader structural pressures shaping urban service delivery [74, 82]. Within this context, settlement morphology characterized by high density, irregular layouts, narrow access routes, poor drainage, and tenure insecurity constrains the technical feasibility of extending safely managed water and sanitation infrastructure [41, 64]. These spatial constraints intersect with demographic dynamics such as large household sizes, tenancy arrangements, migration, low income, and gendered dependency structures, which intensify demand and reduce affordability, thereby reinforcing reliance on shared or informal services [14, 27]. Underlying governance challenges including institutional fragmentation, weak enforcement, limited accountability, and poor neighbourhood-level data systems (Fig. 1), further mediate infrastructure investment, and regulatory oversight, often leaving informal settlements underserved [3, 66]. The convergence of these drivers produces informal WASH service ecologies, including vendor water markets, tanker and sachet dependence, shared or public sanitation, and informal desludging, which, while adaptive, frequently impose higher per-unit costs, unreliable supply, and increased contamination risks on poorer households [15, 58], culminating in entrenched WASH inequality outcomes. These comprise of affordability burdens, sanitation overcrowding, disease exposure, and social exclusion that generate feedback loops of poor health, income loss, infrastructure decline, and data paucity that perpetuate socio-spatial disparities.

3 Methodology

3.1 Study site

The two study sites are neighbourhood communities situated within the Greater Accra Metropolitan Area (GAMA), bordering Accra, the capital city of Ghana Chorkor is located at 5° 31′ 41″ N: 0° 14′ 60″ W (Fig. 2) in the Ablekuma South Municipality and is predominantly inhabited by the indigenous Ga ethnic group. Chorkor is a densely settled coastal community within GAMA having large household sizes, averaging five to seven members, and most dwellings are compound houses constructed with limited sanitation infrastructure [32]. Fishing and fish processing, particularly smoking and drying, are the main economic activities, with women forming a sizeable portion of the labour force in fish marketing and processing [56]. Seasonal variations in fish catch heavily influence household income levels, contributing to fluctuating livelihoods and high poverty incidence [9].

Nima, by contrast, is located at 5° 34′ 55″ N: 0° 11′ 56″ W (Fig. 2) in the Ayawaso East Municipal Assembly, where it also serves as the municipal capital. The settlement has a strong migrant presence, particularly from northern Ghana and neighbouring West African countries [67]. Average household size ranges from six to eight members, reflecting extended family arrangements and high tenancy rates [32]. Unlike Chorkor, the local economy is dominated by trading and small-scale services, including retail markets, transport-related activities, and informal street vending [52]. Nima's inland, high-density settlement structure and complex cultural diversity create different social dynamics compared to the coastal, largely homogenous Chorkor community. These contrasting characteristics informed the selection of the two sites to capture differences in spatial gradient of coastal indigenous versus inland migrant-dominated urban low-income contexts.

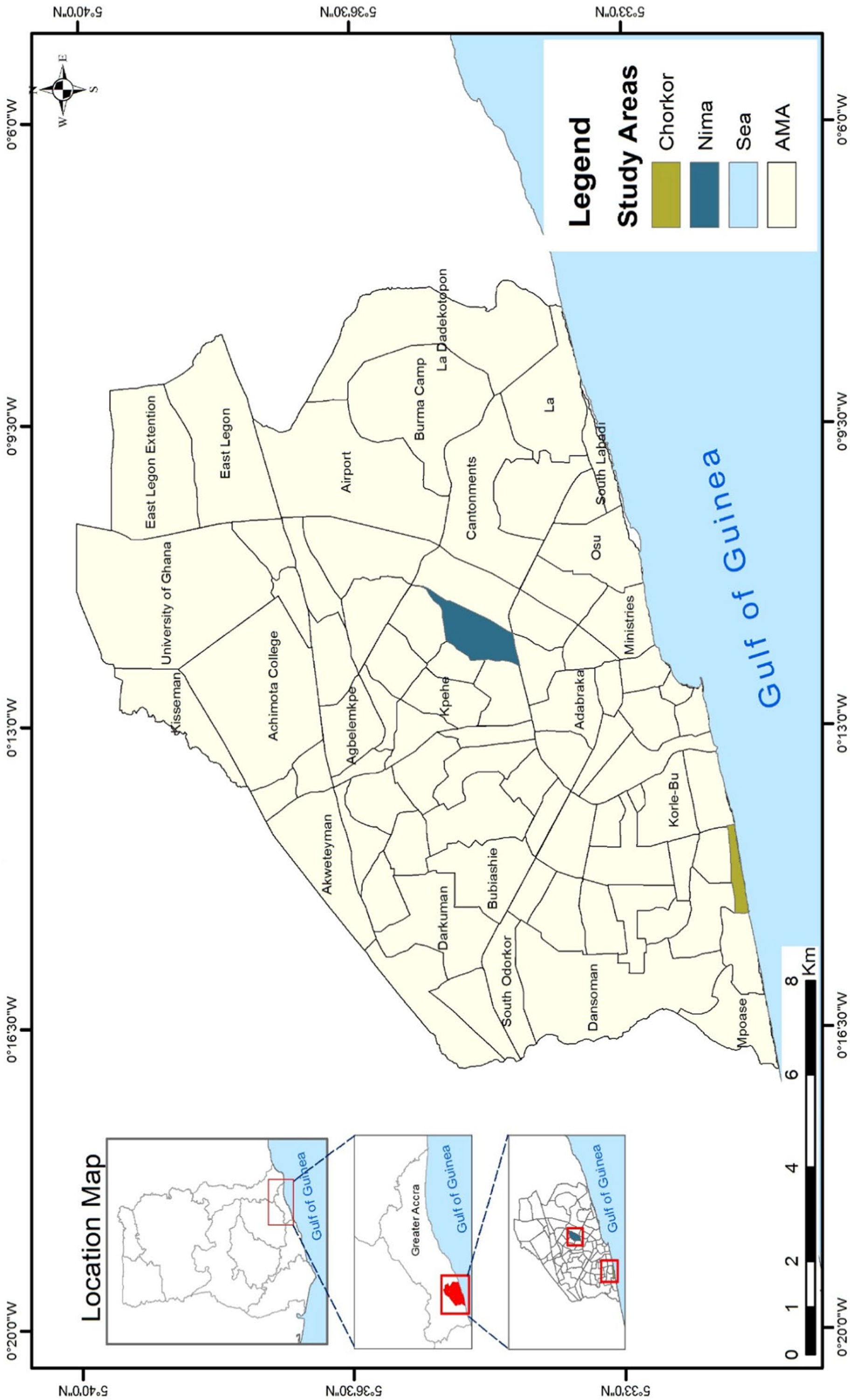

Fig. 2 Map of Accra metropolitan area

3.2 Research design

This study adopted a case study design to enable a nuanced exploration of access to water and sanitation services using multiple sources of evidence [21]. The design supported a mixed methods approach, integrating both quantitative and qualitative techniques to enhance the robustness of findings. Data collection instruments included a semi-structured household questionnaire, in-depth interviews, focus group discussions, and direct observation. Additionally, GPS mapping was used to geolocate households, water sources, toilet facilities, and dumpsites, enhancing the spatial understanding of service access. Primary data was collected using household surveys and qualitative interviews of fifteen key informants alongside focus group discussions (FGD). Secondary data, including annual reports, journal articles, and official records, provided theoretical grounding and context. Prior to the data collection, an introductory letter from the University of Ghana was granted to seek permission from opinion/community leaders at Chorkor and Nima, to conduct the study. A pretest/pilot study was conducted on the instruments guiding the interviews and focus groups that fine-tuned the data collection tools. This was conducted in anticipation of challenges in the actual fieldwork and subsequently shaping the instruments and interview guide. A week's prior notice was given for conducting the interviews. Each member of the sample population was interviewed and copies of the questionnaire distributed to randomly selected households to complete. Key informants were selected purposively for their roles in community-based organizations and local governance. The research was carried out following the guidelines of the research ethics policy of the Ethics Committee for the Humanities (ECH) University of Ghana. The guidelines include respondent safety and privacy measures, voluntary participation and right to withdraw at any time, and how their data would be used, trauma-informed interviewing techniques, and safeguarding principles to mitigate emotional distress or re-traumatization.

3.2.1 Quantitative survey

Household surveys sought to cover household access to water and sanitation and the barriers affecting access to water and sanitation service in the communities was administered. The instruments (Supplementary Information 01) were made up of structured and semi-structured questionnaires and formed the main pathway for seeking responses from households. The household refers to a person or a group of persons, living in the same house or compound and share the same housekeeping arrangements [31]. The simple random sampling technique was employed in administering the household questionnaire. A total households of 150 households made up of seventy-eight from Chorkor and seventy-two from Nima were randomly selected, to minimize selection bias. The target member of each household was the household head, in the absence of which any adult aged 18 years and above within the same household was interviewed.

3.2.2 Focus group discussion

Four (4) focus group discussions, with two in each locality, were conducted. Participants were grouped on the basis of age and sex. Each group consisted of 6–8 persons of the same sex and purposively selected for persons aged between 25 and 60 who have lived in the community for 10 years and above. The 10 years benchmark was used for selecting focus group participants on defining the extent of knowledge and experience of the people about their communities and better positioned to account for past water and sanitation situation in the communities. The participants were selected in the process of administering the household survey, to seek for collective diversity of group opinions at the community level and detailed qualitative data not clarified when administering the survey. The discussions were broadly cantered around the following two topical areas:

i. Water and sanitation services (source of drinking water, type of toilet facility, solid waste stored, household dispose of its solid waste main).
ii. Barriers to effective water and sanitation service (barriers to effective water and sanitation service, problems encountered in accessing water and sanitation, cultural or religious beliefs, programs in your community to improve water and sanitation service).

3.3 Data description and analytical framework

The study followed a structured data collection procedure to examine household access to water and sanitation in Chorkor and Nima. Preliminary steps included obtaining institutional permission through an introductory letter and conducting a pilot study to refine the questionnaire and interview guide, ensuring contextual clarity and validity [21]. Primary data was gathered through household surveys, interviews, and focus group discussions, with prior notice given to key informants. Data were coded and cleaned for analysis, maintaining adherence to ethical and academic standards (i.e. All participants were informed about the purpose of the study, its voluntary nature, and the confidentiality of their responses. Verbal consent was obtained before participation. Data collected were anonymised using codes to ensure that individual participants could not be identified. Personal identifiers were removed during transcription, and recordings were securely stored. The study posed no risks to participants. While no direct benefits were provided, participants contributed to a process aimed at strengthening WASH preparedness and awareness.

Three dependent variables were assessed (access to water, toilet, and solid waste facilities), classified as either improved or unimproved, following global health standards. Improved water sources included piped water and public standpipes, while flush toilets were considered the safest form of improved sanitation. Co-production defines a concrete bundle of community actions and represents the practical forms of cooperation reported by residents, including attitudinal change, communal labour, and education as dominant modes, and providing bins, supporting strict law enforcement, house-owner construction of toilets, more skips and regular emptying, and sanctioning/penalties (fines, community sweeping) suggested in focus groups. So, co-production is applied as everyday joint management and joint maintenance of WASH services, coupled with social regulation and community mobilisation. The independent variables in the model included distance to facility, reliability, education, income, and facility sharing, consistent with findings in existing literature on water and sanitation accessibility [80]. Barriers to access were grouped into environmental, institutional, socio-economic, and attitudinal categories. Quantitative data were analysed using SPSS, Excel, from which Chi-square Tests and T-test techniques to test for associations and group differences were applied. Spatial data collected with GPS were processed in ArcGIS to produce maps depicting service access across communities. Qualitative data, including interview transcripts and field photos, were thematically analysed to complement the statistical findings and provide nuanced insights into lived experiences and infrastructure conditions.

3.4 Weighting access to water and sanitation indicators

WASH Access is defined operationally as the degree to which a household can obtain and use water and sanitation services in ways that meet acceptable thresholds. Household access was measured using a set of indicators drawn from WASH and urban services literature and validated through focus group discussions, key informant interviews, and expert judgment. The indicators were economic, spatial, technical, and functional dimensions of WASH access (Table 1). Each indicator was assigned a score ranging from 0 to 10, where: ten represents the optimum access condition, and zero represents the least desirable or most constrained condition. Intermediate scores reflect graduated levels of access (e.g., partial affordability, moderate distance, intermittent reliability). Scoring thresholds were established using participatory inputs from community focus groups, key informants, and expert opinion, informed by existing empirical studies which ranged from zero to fifty (0–50). These were grouped into (i) Access, (ii) Partial Access and (iii) No Access, when a total score of (i) 40–50, (ii) 39–20 and (iii) below 20,

Table 1 Weighting indicators for access to water and sanitation

Indicators	Descriptions and scores			
Income	Spending 3% of income and below	4–6%	7–10%	> 10%
	10	7	5	0
Distance	Source in dwelling	Travelling < 1000 m	Travelling > 1000 m	
	10	5	0	
Source	Pipe in dwelling	Public tap	Borehole	Water tankers
	10	5	3	0
Reliability	Very reliable	Reliable	A few times	Not reliable
	10	5	3	0
Quantity	> 40 L daily	40–20 L daily	< 20 L daily	
	10	5	0	
Sharing of toilet facility	Exclusive use	Sharing but not public	Public Toilet	
	10	5	0	
Usability	Flush toilet	Pit latrine with slab or Ventilated Improved Pit latrine	Pit latrine with no slab	Bucket latrines/Open Defecation
	10	5	3	**0**
Waste disposal	Door-to-door/skips	Other means		
	10	0		
Containment	Safe containers protected from flies	No container		
	10	0		

respectively (Table 1). For each household, an overall Access Score was computed by summing the individual indicator scores:

$$Access\ Score_h = \sum_{i=1}^{n} S_{hi}$$

where S_{hi} is the score assigned to household h on indicator i. The resulting composite score ranges from 0 to 50.

4 Findings

4.1 Source and related distance of access to water

In Chorkor, certain zones particularly those nearer to coastal roads or key infrastructure demonstrate relatively higher access (Fig. 3). Other segments particularly those farther from the main access roads, suffer from moderate to low water availability, pointing to spatial inequities in service distribution. Conversely, Nima shows a more uniformly low to moderate level of access. Households across much of the community reported consistently limited access to water. Unlike Chorkor, Nima does not exhibit high-access clusters, indicating a widespread infrastructure inadequacy rather than a spatially unequal one (Fig. 3). Most households in both communities' access water within their community rather than in their dwellings (Chorkor: 89.7%, Nima: 86.1%). Despite minor differences (10.3% in-dwelling in Chorkor vs. 13.9% in Nima), the p-value of 0.494 indicates no significant difference. This reflects a shared infrastructure reality where most residents depend on communal water points. However, there is no differences in the distribution of water sources at Chorkor and Nima ($p > 0.05$).

The results show that at Chorkor, about 30.8% of households travel less than 200 m to a water point, 48.7% also travel between 200 and 1000 m and the remaining 20.5% travel more than 1000 m to access water. At Nima, on the other hand, 48.6% travel less than 200 m to access water while 38.9% travel a distance between 200–1000 m. The remaining 12.5% travel beyond 1000 m to access water. The mean distribution of distance to a water source at Chorkor and Nima shows that the differences in the distance travelled to water sources between and within Chorkor and Nima is statistically significant ($p < 0.05$). Overall, Chorkor residents travel a mean distance of 69.49 m, while Nima residents travel slightly less (54.93 m). The pooled mean across both communities is 62.5 m. These results highlight a structural inequity where Chorkor households face longer walks and underscoring the physical burden of water collection.

4.2 Access to sanitation facilities

More than half of households at Chorkor and Nima, utilize public toilets in the community, whereas a quarter of households in both communities have individual household toilets for their exclusive use. The percentage distribution of the types of toilet facility shows that the differences in the types of toilet facilities at Chorkor and Nima is not statistically significant ($p > 0.05$). The proportion of households using the improved type (flush toilet) of toilet facilities is 36% and 32% at Chorkor and Nima, respectively, while the remaining use the unimproved types of facilities such as pit latrines with/without slabs, ventilated improved pit latrine, bucket/pan latrines, and open defecation.

Fig. 3 Location and access to water at Chorkor and Nima, GAMA, Ghana

Although the distribution of facilities varies, the differences are statistically insignificant ($p = 0.145$). In Chorkor, flush toilets (35.9%) and VIP latrines (30.8%) dominate, whereas Nima shows higher reliance on pit latrines without slabs (27.8%). Open defecation is marginal in both contexts (Chorkor: 5.1%, Nima: 4.2%). While usage patterns differ, both communities reflect a reliance on substandard facilities, with improved toilets remaining limited. The percentage distribution of the sharing of toilet facility at Chorkor and Nima shows that the differences in the sharing of toilet facility in both communities is not statistically significant ($p > 0.05$). Sharing patterns are identical, with two-thirds of respondents in both areas relying on public toilets (Chorkor: 62.8%, Nima: 63.9%). Individual ownership remains low (20.5% in Chorkor, 13.9% in Nima). The non-significant p-value of 0.362 highlights a common dependence on communal sanitation, reinforcing challenges of overcrowding and hygiene.

4.3 Differences in access to water and sanitation

Significant differences were registered in access to improved water facilities ($p = 0.017$). Chorkor fares better (75% improved) compared to Nima (56.4%). However, toilet access remains unimproved in both communities, with no significant difference (Chorkor: 87.2% unimproved; Nima: 91.7%). In terms of types of access (access, partial access, no access), household level access to water (Fig. 3) shows that 33% of sampled households at Chorkor have no access to water as against 25% at Nima. An estimated 35% of the sampled population have partial access to water at Chorkor and 24% at Nima. At Chorkor, 32% of respondents have access to water as against 51% at Nima. Solid waste facilities showed non-significant difference ($p = 0.057$), with Nima (48.6%) reporting better access than Chorkor (33.3%) (Table 2).

The proportion of household with no access to toilet facilities at Chorkor was 27% compared to 26% at Nima (Fig. 4). In the partial access category, Chorkor recorded 32% as against 46% at Nima. For toilet facilities at Chorkor, 42% of households have access, compared to 28% at Nima. For access to solid waste disposal facilities, the proportion

Table 2 Access to water and sanitation in the communities

Access to water facilities	Improved[a] (%)	Unimproved[b] (%)	p-value
Chorkor	75	25	0.017
Nima	56.4	43.6	
Access to toilet facilities	**Improved[aa] (%)**	**Unimproved[bb] (%)**	**p-value**
Chorkor	12.8	87.2	0.374
Nima	8.3	91.7	
Access to solid waste facilities	**Improved[aaa] (%)**	**Unimproved[bbb] (%)**	**p-value**
Chorkor	33.3	66.7	0.057
Nima	48.6	51.4	

[a] Piped water in dwelling, public tap or standpipe,

[aa] Flush toilet

[aaa] Skips, door-to-door collection

[b] Sachet, bottled water, Borehole, Water tanker, Protected dug well, others

[bb] Ventilated improved pit, Pit latrine with slab, Pit latrine without slab, Bucket/Pan latrine, others

[bbb] Drains, sea, lagoon

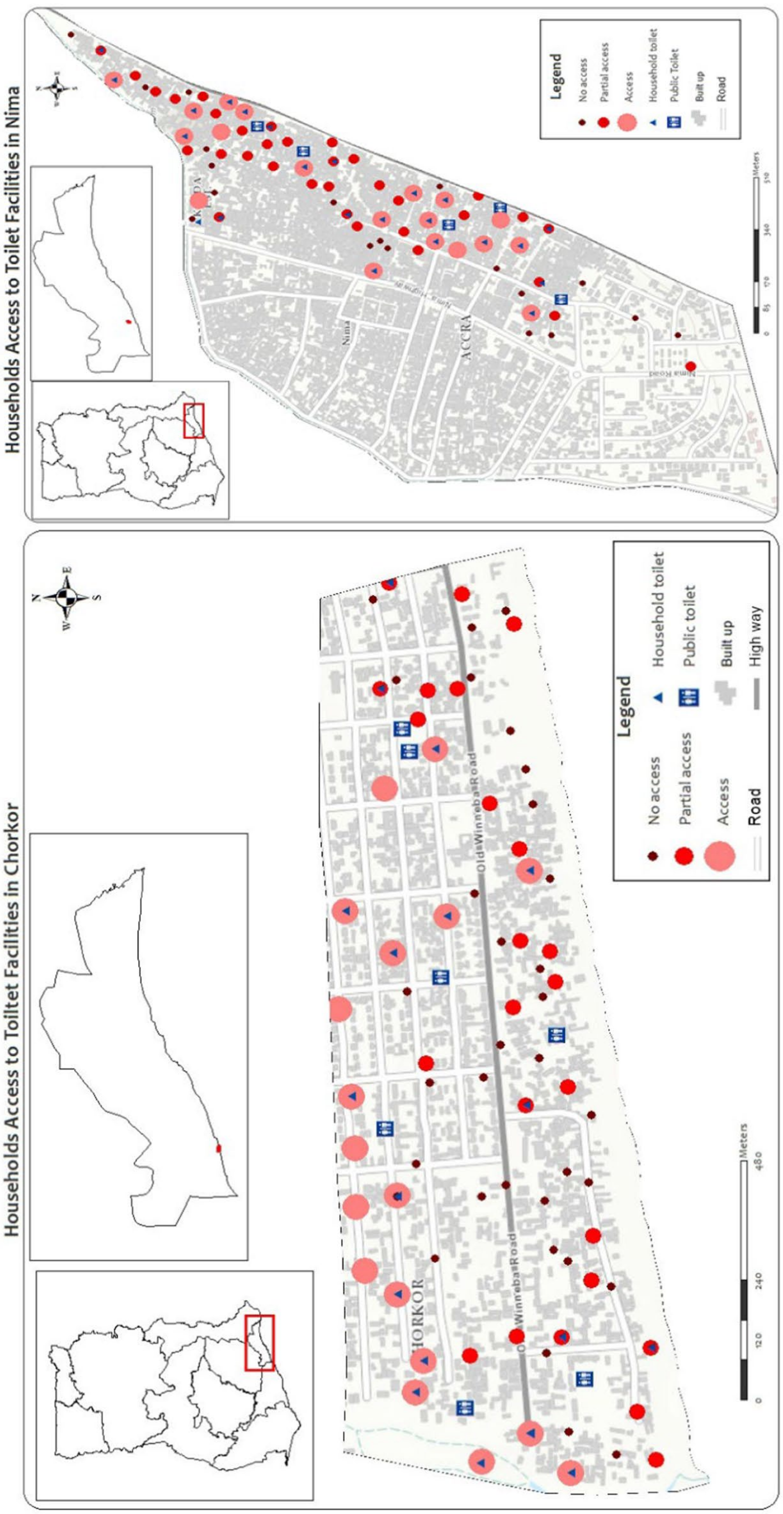

Fig. 4 Location of, and access to toilet facilities at Chorkor and Nima, GAMA, Ghana

of household without access, stood at 28% of households at Chorkor compared to solid waste disposal facilities of 35% at Nima. In the partial access category, Chorkor recorded 35% as against 49% for Nima. The proportion of households with access to solid waste disposal facilities at Chorkor stood at 37% as against 16% for Nima. Solid waste disposal infrastructure is crucial for maintaining environmental health, yet it remains underdeveloped in many urban slums. Chorkor displays a fragmented pattern (Fig. 5), where access to solid waste disposal points is inconsistent and particularly poor in coastal and densely populated areas. In contrast, Nima shows a more even distribution of access to solid waste disposal points, especially along the main thoroughfares and commercial routes (Fig. 5).

4.4 Intersectionality of Access to Water, Sanitation against Households' Attributes

The degree of association between access to improved water and sanitation and selected variables are reported (Table 3). Sociodemographic factors show association with access to water, toilet, and solid waste facilities in both communities ($p < 0.05$). Thus, access to drinking water is strongly influenced by income and distance in both communities ($p < 0.001$ in Chorkor; $p < 0.05$ in Nima), whilst education is not a significant determinant. Access to toilets is income-dependent in Chorkor ($p = 0.024$) but not in Nima. In both communities, distance significantly determines access (Table 3). Sharing facilities is significant only in Chorkor ($p = 0.000$), as solid waste disposal shows strong associations with education and distance in Chorkor, and only distance being significant in Nima (Table 3).

Several statistically significant contrasts appear across the various barriers as distance and location are reported by many but are not statistically different (Table 4). However, lack of access roads is significant ($p = 0.000$), with Nima more affected compared to Chorkor. Attitude and maintenance culture significantly differ, with Nima residents reporting more positive practices ($p = 0.010$; $p = 0.032$) (Table 4). Whilst lack of consultation is significantly worse in Nima ($p = 0.000$), there was no difference in the weak law enforcement in both communities. In addition, income and education are highly significant ($p = 0.000$), with Chorkor residents reporting greater challenges.

4.5 Community support to WASH facilities (community co-production)

The contributions of communities to WASH improvement significantly affect access in Nima but not in Chorkor. In Nima, household/community efforts improved access to water facilities ($p = 0.002$), toilet facilities ($p = 0.047$), and waste management facilities ($p = 0.002$). Chorkor, however, shows no significant effect of any of the service facilities. Residents of both communities help improve water and sanitation effects through attitudinal change, communal labour and education (Fig. 6). Other measures were the provision of bins, strict law enforcement, construction of toilets in homes by house owners, provision of more skips and regular emptying of skips. Participants of the focus group discussion also suggested stiffer punishment to deter people from engaging in unacceptable sanitation practices. Punishment such as fines and sweeping of public places were suggested to deter people who flout environmental sanitation regulations.

Fig. 5 Location of, and access to toilet facilities at Chorkor and Nima, GAMA, Ghana

Table 3 Association between water and sanitation and some selected variables

Dependent variables	Independent variables	Chorkor		Nima	
		Pearson chi square	*p*-value	Pearson chi square	*p*-value
Access to drinking water	Education	2.786	0.733	6.543	0.162
	Income	18.092	0.000***	8.275	0.016**
	Distance	22.565	0.000***	19.074	0.000***
	Quantity	10.670	0.557	25.905	0.027**
Access to toilet facility	Education	6.211	0.286	5.814	0.213
	Income	7.466	0.024**	1.074	0.585
	Distance	15.214	0.002***	11.520	0.009***
	Sharing facility	27.132	0.000***	1.110	1.574
Solid waste disposal	Education	23.324	0.000***	2.879	0.578
	Income	1.161	0.560	1.566	0.457
	Distance	17.263	0.001***	9.153	0.027**

* $p < 0.05$ is significant; ** $p < 0.01$ is very significant; *** $p < 0.001$ is very highly significant

Table 4 Barriers affecting access to water and sanitation at Chorkor and Nima

		Level of agreement with barrier						*p*-value
		Chorkor			Nima			
		Agree	Neutral	Disagree	Agree	Neutral	Disagree	
Environmental barriers	Location	46.2	10.3	43.5	55.6	12.5	31.9	0.340
	Distance	78.2	9	12.8	66.7	8.3	25	0.159
	Lack of access roads	14.1	27	59	54.2	26.4	19.4	0.000***
Cultural and attitudinal barriers	Attitude and behaviour	84.6	11.5	3.8	72.2	8.3	19.4	0.010**
	Maintenance culture	53.8	32.1	14.1	68.1	13.9	18.1	0.032*
Institutional barriers	Weak law enforcement	69.2	19.2	10.3	84.7	6.9	8.3	0.062
	Lack of local consultation	53.8	32.1	14.1	38.9	13.9	47.2	0.000***
Socio-economic barriers	Income	79.5	14.1	6.4	66.7	4.2	29.2	0.000***
	Educational Level	69.2	15.4	15.4	45.8	9.7	44.4	0.000***

* $p < 0.05$ is significant

** $p < 0.01$ is very significant

*** $p < 0.001$ is very highly significant

5 Discussion

5.1 Household level access to water and sanitation

The structural distribution of access to water in Chorkor along key public infrastructure suggests enhanced access and benefits from informal water vendors (Table 5), facilitated by proximity to tanker supply routes and intermittent piped connections. This pattern reflects a patchwork infrastructure, shaped by localized and informal arrangements rather than comprehensive WASH network coverage (Fig. 3). In contrast, the spatial configuration of Nima depicts water scarcity as a collective condition affecting the settlement broadly, rather than being confined to peripheral or poorly connected zones (Table 5). This divergence implies that while Chorkor may benefit from spatially targeted

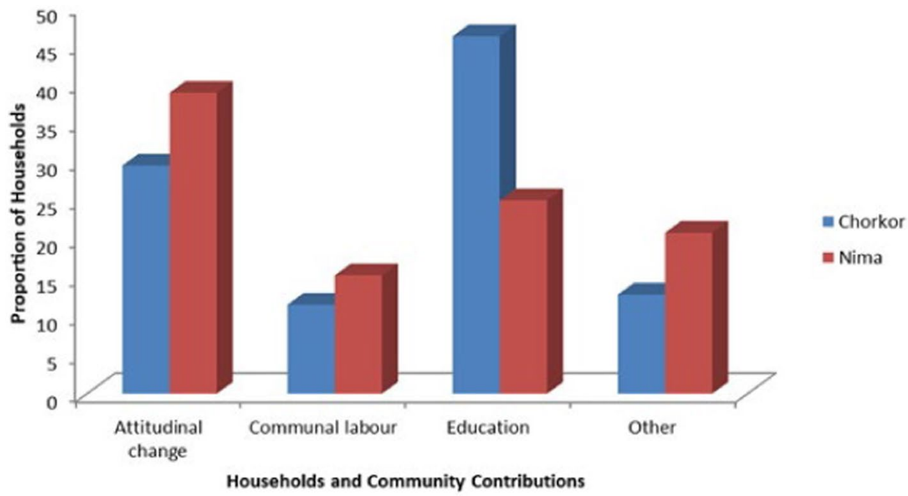

Fig. 6 Modes by which households contribute to enhancing water and sanitation

interventions to address internal inequities, Nima requires community-wide infrastructure upgrading to raise overall service standards. Across both communities, the majority of households rely on communal water sources, which include public standpipes and kiosks. Similar reliance on community-based sources has been documented in informal settlements in Lilongwe and Lusaka, where the absence of household-level piped infrastructure renders water supply unaffordable, unreliable, and unsafe, exacerbating household vulnerability [6]. Such conditions entrench cycles of infrastructure exclusion, in which informal systems emerge as costly and time-consuming substitutes for formal provision. Comparable dynamics are observed in Mumbai's Dharavi slum, where fewer than 30% of households have direct piped connections, and access is instead governed by scheduled water trucks, shared standpipes, and local power relations [57].

Distance to water sources remains a critical barrier to equitable access. Average fetching distance though seemingly modest, translate into cumulative time burdens, particularly for women and children. Such spatial and temporal burdens are widely recognized as hallmarks of water insecurity in informal urban contexts [6]. Evidence from Beira, Mozambique, shows that intra-community distance to water infrastructure affects both access and pricing, disproportionately disadvantaging households located farther from service nodes [77]. Similar patterns are reported across Lusaka, Dhaka, and Mumbai where in-house connections remain a privilege for a minority [47]. In Nairobi's Mukuru settlement, for example, approximately 80% of households rely on informal vendors, leading to inflated prices and rationing during dry seasons [49]. These patterns reflect broader policy failures in LMICs, where informal settlements are frequently excluded from formal water provision due to insecure tenure, spatial illegality, and infrastructure deficits [23, 66].

The predominance of shared and unimproved sanitation in both study communities reflect entrenched structural conditions which are characteristic of dense informal settlements globally, where infrastructure provision has failed to keep pace with population growth [43]. The high reliance on public toilets signals not only limited household-level investment capacity but also institutional neglect in extending affordable, and private

Table 5 Summary of comparative observations across the WASH dimensions in Chorkor and nima, with targeted policy recommendations

WASH Dimension	Chorkor	Nima	Recommendations/policy Implications
Water access	Highly uneven across the settlement. Some zones have high access, while others are underserved, suggesting localized water supply networks	Uniformly low to moderate access across the area, indicating system-wide deficiencies	Chorkor: Prioritize micro-zonal water infrastructure planning by extending piped networks to underserved coastal and interior clusters; regulate and formalize informal water vendors to reduce price inequities Nima: Implement community-wide water system upgrades, including increased standpipe density and reliability improvements, rather than selective extensions
Toilet Access	Predominantly shared facilities; many areas lack private household toilets, especially near the coast	Also reliant on shared toilets, with slightly more private access in select clusters	Chorkor: Strengthen household and compound-level sanitation through targeted subsidies, landlord incentives, and enforcement of minimum sanitation standards in coastal zones Nima: Upgrade shared sanitation facilities with improved maintenance, gender-sensitive design, and incremental conversion to semi-private facilities in high-density clusters
Waste disposal access	Fragmented and insufficient, particularly in coastal and peripheral regions	Relatively better distributed, especially along main routes and commercial zones	Chorkor: Expand solid waste infrastructure coverage to peripheral and coastal areas, including additional skips and scheduled collection, and strengthen enforcement against shoreline dumping Nima: Consolidate existing waste management gains by improving service regularity and integrating community-led waste monitoring along secondary streets
Overall pattern	Access is spatially segregated; some parts fare well while others are excluded. Infrastructure provision is inconsistent	Access is consistently low to moderate, indicating generalized infrastructure inadequacy	Chorkor: Adopt spatial justice-oriented planning that explicitly targets neglected zones through place-based WASH investments Nima: Shift from piecemeal interventions to settlement-wide infrastructure upgrading, aligned with participatory planning and co-production frameworks

sanitation options. Such sanitation regimes disproportionately burden women, children, and older persons, for whom distance, queuing, safety, and privacy constraints translate into heightened health and psychosocial risks. Spatial concentration of poor sanitation in Chorkor (Fig. 4), illustrates how settlement morphology and environmental exposure intersect to intensify contamination and disease risks, while also reinforcing dependence on overcrowded and poorly maintained shared facilities. Although Nima similarly depends on shared sanitation, the spatial analysis indicates marginally better distribution patterns, suggesting the influence of localized infrastructure clustering and community-level interventions. Comparable conditions have been documented in Mumbai's Dharavi, where chronic underinvestment results in overburdened shared toilets that remain the default sanitation solution for most residents [47]. Across contexts, shared sanitation occupies an ambiguous position and that, while preferable to open defecation, it frequently perpetuates undignified, unsafe, and unhealthy conditions due to overcrowding, poor maintenance, and limited oversight [43]. Elsewhere, a gender-sensitive lens reveals how sanitation insecurity exposes women and girls to physical and psychological harm particularly where facilities lack lighting, privacy, or safe access routes, in Bangladesh and Nairobi's Kibera settlement [20, 70]. These dynamics underscore that sanitation outcomes are shaped as much by social relations and safety considerations as by physical infrastructure.

Socioeconomic stratification further explains the observed access patterns. In Chorkor, income emerges as a primary determinant of toilet access, reflecting affordability constraints and uneven household investment capacity. In contrast, Nima's sanitation outcomes are more strongly shaped by spatial location and institutional arrangements, including integration with municipal service networks. Similar patterns have been reported in Lusaka and Blantyre, where socioeconomic disparities and governance failures jointly entrench exclusion [6]. Spatial dispersion of waste facilities in Chorkor highlights planning and logistical deficits, whereas relatively cohesive coverage in Nima suggests more coordinated service delivery. These findings mirror global evidence that sanitation inequalities persist despite SDG progress, driven by interacting effects of income, geography, governance capacity, gender, and education across the Global South [12, 22, 48, 51, 60, 63].

5.2 Limiting access to water and sanitation, improvement and co-production

The findings reveal two distinct patterns of urban WASH exclusion. In Nima, limited access to water and sanitation is relatively uniform across the settlement, indicating systemic under-service, whereas Chorkor exhibits internal spatial differentiation, with certain areas enjoying better access due to historical planning decisions and proximity to arterial roads and trunk infrastructure. Comparable settlement-level inequalities have been documented in Latin America, where socio-economic segregation and neighbourhood spatial configuration strongly shape sanitation access [48, 83]. These patterns reflect broader urban development dynamics in which infrastructure expansion follows formal planning logics, often bypassing informal or peripheral zones. In Chorkor, income emerges as a statistically significant determinant of toilet access, suggesting that even nominally public sanitation options entail costs or require private investment in household latrines. Similar income-mediated inequalities have been widely observed across Latin America and Asia, where affordability constrains effective sanitation access despite physical availability [33, 60]. Distance and location further exacerbate exclusion, as households situated far from trunk infrastructure or access roads are frequently omitted from service grids, as reported in Dar es Salaam [8]. Parallel patterns in South Asia demonstrate that peripheral and marginalized districts face persistent barriers to improved water and sanitation [18, 22]. While Chorkor benefits unevenly from infrastructure proximity, Nima's uniformly low access suggests entrenched neglect, where even central-city location does not translate into service provision, an outcome that is consistent with governance failures reported across Latin America and the wider Global South [34].

From a governance perspective, the findings support arguments that market-oriented service delivery without strong pro-poor regulation deepens inequality [28]. In Nima, weak enforcement and limited community consultation have resulted in reactive rather than anticipatory WASH interventions, reflecting critiques of top-down urban planning systems that fail to engage informal communities [25]. Cross-country evidence highlights a persistent mismatch between formal planning frameworks and the lived realities of informal settlements, leading to systematic exclusion [66]. In Ghana, fragmented institutional mandates and weak inter-agency coordination mirror challenges observed in cities such as Maputo and Medellín, where overlapping jurisdictions and limited participation perpetuate inequitable WASH outcomes [25]. Even where aggregate urban coverage appears high, localized exclusions remain hidden, underscoring the value of spatially disaggregated analysis [36]. Despite these structural constraints, the findings also highlight community resilience, particularly in Nima. Community-based organizations have supported public toilet maintenance, waste management, and advocacy, exemplifying co-production in WASH service delivery. However, in Lusaka and Maputo, fragmented governance and weak institutional coordination perpetuate inequalities in WASH access despite national urban development policies [25, 44]. Evidence from diverse contexts suggests that such community-driven arrangements can enhance sustainability where state provision is weak [47, 50]. In contrast, Chorkor's deeper poverty and lower educational attainment constrain collective agency, limiting the effectiveness of co-production [6]. These contrasts underscore the practical implication that WASH interventions must be context-specific, recognizing how socio-economic capacity and institutional responsiveness condition community participation. Thus, informal settlements in Pakistan and South Africa shows that community-managed sanitation and waste systems can improve service sustainability where formal systems fail [10, 17, 65]. The study thus emphasizes inclusive urban planning, participatory governance, and equitable access to urban services in the Global South with the lens of human rights for informed access that is available, accessible, safe, acceptable, and affordable for all without discrimination [75, 78].

In Dhaka, informal settlements experience chronic water shortages and sanitation congestion despite the city's extensive water infrastructure, as migrant populations living in low-income settlements often pay significantly higher prices for water supplied by informal intermediaries than wealthier households [37]. Likewise, in Karachi, tanker-water economies dominate peri-urban informal communities where weak governance and illegal land occupation prevent formal service expansion, creating highly unequal patterns of water access linked to class and political influence [35]. Similarly, in Rio de Janeiro, favela communities reveal another dimension of WASH inequality where infrastructural improvements coexist with social exclusion. While some settlements have piped water, sanitation services remain unreliable because of fragmented governance, informal housing structures, and violence-related barriers to municipal access [59]. Similar conditions are observed in Lima, where peripheral desert settlements depend on expensive trucked water despite being located near one of Latin America's largest metropolitan economies. Poorer households in Lima may spend proportionally far more on water than affluent urban residents connected to formal systems [40]. Across the Middle East and North Africa, such as in Cairo, informal urban expansion has produced sanitation overloads and drainage failures in densely populated settlements, particularly affecting migrant and low-income households lacking secure tenure [61]. In Cape Town, the unequal geography of apartheid-era planning continues to shape access to sanitation and water reliability in Black townships and informal settlements despite post-apartheid reforms [24]. Small island and climate-vulnerable urban contexts provide additional parallels as observed in Port-au-Prince in Haiti, where weak institutions, vulnerability to disasters, and poverty intersect to undermine sanitation and waste management systems, particularly after the 2010 earthquake and cholera outbreak [26]. Similarly, in Jakarta, flooding, groundwater extraction, and informal urbanisation mutually impose severe inequalities in sanitation and clean water access, disproportionately affecting coastal informal settlements [19].

6 Conclusion

Both study communities face significant challenges, yet their spatial characteristics differ along an infrastructure evenness, segmentation, and uniformity with distinct implications for WASH interventions and improvements. Revealing these contrasting intra-urban "regimes of access" which is spatially segmented service provision in Chorkor against settlement-wide systemic under-service in Nima, the study goes beyond conventional portrayals of informal settlements as uniformly deprived and demonstrates that WASH inequality is structurally patterned within low-income urban space. Understanding these spatial nuances is essential for policymakers, urban planners, and NGOs aiming to improve urban health equity. There is critical need for tailored approaches that consider spatial dynamics, community needs, and resource constraints for achieving sustainable WASH outcomes in these vulnerable communities. The study's integrative use of spatial mapping, multidimensional access indicators, and household-level analysis adds empirical precision to this need by showing that "access" is not merely about infrastructure presence, but about effective serviceability shaped by affordability, distance, reliability, and sharing burdens. Understanding these nuances is essential for policymakers, urban planners, and NGOs aiming to improve urban health equity. Such nuances further reveal differences in barriers faced by the communities in aspects of institutions (i.e., consultation, roads), and socio-economic dynamics (i.e., income, education). It reinforces the global reality that while WASH is recognized as basic rights and development priorities, delivery remains stratified by income, geography, and prudent governance. Within this landscape of exclusion, there is also evidence of local agency and resilience through community co-production when supported by responsive institutions, to deliver tangible improvements in WASH access. As such, policies aimed at improving water and sanitation in low-income urban communities must embrace a pluralistic model, one that integrates spatial justice, participatory governance, and socio-economic inclusion. Looking into future research, scaling the analysis to engage spatial statistics in analysing distinct spatial exclusion regimes will complement other approaches towards enhanced empirical basis for linking WASH access to broader development pathways including the SDGs.

Supplementary Information

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Supplementary Material 1.

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GAO developed the concept and collected the data, Both DBKD and GAO wrote the initial manuscript text including analysing the data, and MK and AKC provided a critical review and further rewriting of the manuscript which was updated and supervised by DBKD.

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Declarations

Ethics approval and consent to participate
The authors submitted survey protocol to the University of Ghana before the research, followed the research ethics policy of the Ethics Committee for the Humanities (ECH) University of Ghana confirm no formal approval is needed. All the terms have been clearly stated in the survey questionnaire, so all the participants participated in the survey voluntarily with their Informed Consent. Informed consent to participate from households and subjects was sought.

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